

International Journal of Food Sciences and Nutrition

ISSN: 0963-7486 (Print) 1465-3478 (Online) Journal homepage: https://www.tandfonline.com/loi/iijf20

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To cite this article: E. Salvucci (2019): The human-microbiome superorganism and its modulation to restore health, International Journal of Food Sciences and Nutrition, DOI: 10.1080/09637486.2019.1580682

To link to this article: https://doi.org/10.1080/09637486.2019.1580682

Published online: 07 Mar 2019.

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INTERNATIONAL JOURNAL OF FOOD SCIENCES AND NUTRITION
https://doi.org/10.1080/09637486.2019.1580682

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COMPREHENSIVE REVIEW

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The human-microbiome superorganism and its modulation to restore health

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ABSTRACT

Microbiome is the community of microorganism that co-live with a host. The human being is the result of the integration of its genome and the coexistence with millions of microorganisms throughout its evolutionary history. Human-microbiome association can be considered a step of integration in evolution, constituting a *superorganism*. Many emergent diseases are related to the loss of part of this microbiome and its restoration can be achieved by different strategies. Gut microbiome imbalance is particularly associated with numerous inflammatory, immune and nervous system-related diseases by a communication pathway called microbiome–brain axis. Modulation of microbiome by administering prebiotics, like arabinoxylans, and synbiotics is a plausible treatment for dysbiosis, the regulation of neurotransmitters and alleviation of neurological manifestations.

ARTICLE HISTORY
Received 2 November 2018
Revised 5 February 2019
Accepted 6 February 2019

KEYWORDS
Microbiome; prebiotics; probiotics; neurodevelopmental disorders; *Bifidobacterium*; microbiome–brain axis

Introduction

Human microbiome is the full complement of microbes (bacteria, viruses, fungi and protozoa) and their genes in the human host. These associated microorganisms are a constitutive part of the human being (Bakhtiar et al. 2013; Salvucci 2016). The human body is the result of the integration of its genome and the coexistence with millions of microorganisms throughout thousands of years of its evolutionary history. In particular, gut microbiome is a taxonomically complex and ecologically dynamic community that influences the development, maturation and regulation of the immune system (Salvucci 2016; van der Meulen et al. 2016). It is assumed that the number of microbial cells outnumbers human cells in a ratio 10:1. Recently, it was estimated that this ratio is 1.3:1 (Sender et al. 2016) but counting just bacterial cells. Four bacterial phyla are the dominant groups: Firmicutes, Bacteroidetes, Actinobacteria and Proteobacteria. Eukaryotic microorganisms are also part of microbiome. Mycobiome (fungi microbiome) comprise a smaller proportion of human microbial sequences as compared with bacteria and most of them can only be detected with non-culture-based techniques. Nevertheless, it remains an important component of the human microbiome (Cui et al. 2013). Particularly, the gut mycobiome contains species that belong to the genera *Candida*, *Saccharomyces*, *Aspergillus*, *Cryptococcus*, *Malassezia*, *Cladosporium*, *Galactomyces*, *Pentatrichomonas*, *Entamoeba* and *Trichosporon* (Parfrey et al. 2011; Hallen-Adams and Suhr 2017; Nash et al. 2017).

All these microorganisms constitute a complex system whose totality exceeds the sum of the parts, influencing one another in an intricate metabolism. The virome is also part of this complex community. It includes the collection of nucleic acids from viruses. These are involved in horizontal gene exchange contributing to rapid response to environmental changes and bacterial evolution (Jones 2010). Human being could be considered as a step of integration in evolution, constituting a superorganism (Sleator 2010; Kramer and Bressan 2015; Salvucci 2013, 2016).

The human microbiome has contributed to the emergence of the immune system. This is a complex network of cells and signals that was shaped (and emerged) in association with microbes. The immune system is the result of integration of systems. The early precursors of the adaptive immune system emerge around 500 million years ago in some species of non-jawed vertebrates, possibly the result of mobile microbial DNA "infecting" marine creatures (West et al. 2015). The machinery for lymphocyte diversity emerge by recombination activating gene (RAG) homologues (Fugmann 2010). Mechanisms like the

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integration of virus and the variability provided by transposable elements (TEs) combined with regulation of gene expression are involved throughout evolution (Feschotte 2008; Chuong et al. 2017; Colson et al. 2018; Rivas-Carrillo et al. 2018).

Microbiome also constitutes an integral part of fundamental processes such as vitamin production, digestion, energetic homeostasis, integrity of the intestinal barrier and angiogenesis in the human body (Benson 2016). Since the industrial era, there was an increasing incidence of a group of diseases that are characterised by an overactive immune system. In this regard, numerous advances have been made in the understanding of metabolic and immune diseases and their relationship to the microbiome. A better understanding of microbiome as an essential counterpart of the human superorganism has led to find a common background of immune, inflammatory or systemic imbalance in a group of diseases (Salvucci 2016; van der Meulen et al. 2016). For instance, it is reported the connection between increased ratio of main groups Firmicutes:Bacteroidetes and metabolism alterations observed in obesity (Tilg and Kaser 2011; Bakhtiar et al. 2013; Koleva et al. 2015).

Substantial progress has also been made in the study of the relationship of microbiome with allergies and immune diseases. Bacteria (for instance, *Lactobacillus* spp., bifidobacteria) and helminths (like *H. polygyrus*) can modulate and restore an overactive immune system and prevent the development of allergic reactivity. In this process, enhancing of Treg cells and modulation of Th2 response are involved. Susceptibility to allergy development in humans and in animal allergy models can result from either antibiotic or antihelmintic treatment (Mazmanian et al. 2005; Rook 2009; Reynolds et al. 2015).

Diet and nutrition influence the microbiome and has profound effect on systemic equilibrium. Dysbiosis is a term that refers to the alteration or shift in the structure of the microbiome associated with different diseases. It becomes necessary to get a precisely understanding of the changes that occurs in each disease and recognise which microbial communities or specific microbes cause or contribute to the disease (Duvallet et al. 2017). The main changes are related to bacterial community and recent works suggested that also fungal dysbiosis might contribute to the pathology of several immune-mediated conditions of non-infectious origin (Iliev and Leonardi 2017). Several diseases that were formerly considered to have no association with fungi, such as hepatitis B, cystic fibrosis and inflammatory bowel disease (IBD) are now found to be associated with particular mycobiomes (Cui et al. 2013; Iliev and Leonardi 2017). Based on the reciprocal interaction of diet and microbiome it is possible to modulate the functionality of microbiome by diet or components of food. This modulation by administration of prebiotics or synbiotics can be an alternative treatment of co-occurring gut symptoms associated to neurodevelopmental disorders and to alleviate the symptoms of the disease through a systemic effect.

The superorganisms

All organisms, including human, are interconnected in a metabolic network. Evolution and emergence of species are the result of that net (Raoult 2010; Merhej and Raoult 2012; Ramulu et al. 2012). Eukaryotic cells emerge by integration of bacterial cells. Organelles like mitochondria have bacterial origin and do not have a single common ancestor. Most likely, they have numerous ancestors, including proto-Rickettsiales, proto-Rhizobiales and proto-Alphaproteobacteria (Georgiades and Raoult 2011). This is a net of life, an intricate interrelationship between all the organism in nature as was stated by several authors (Jablonka and Raz 2009; Gilbert et al. 2010; Salvucci 2016).

Symbiogenesis proposed independently by Konstantin Mereschkowski, Boris Kozo-Polyanski, Ivan Wallin, Paul Poitier, Lynn Margulis and others, has accumulated strong evidence to be considered a major mechanism in evolution (Margulis and Fester 1991; Kozo-Polyansky and Margulis 2010; Salvucci 2012; Aanen and Eggleton 2017; Lane 2017). It is a process observed at all levels in nature (Margulis and Chapman 1998; Brucker and Bordenstein 2012; Aanen and Eggleton 2017). The integration of symbiont cells resulted in new cell types and also new species (i.e. lichen), promotes reproductive barriers (i.e. *Wolbachia* and arthropods) and hybrid incompatibilities (Brucker and Bordenstein 2012; Aanen and Eggleton 2017). This fundamental process in evolution should be taken into account in order to get a better understanding of human evolution and the health/disease status.

The complex evolutionary relationships of the organisms are revealed in their genomes. Genomes from Archaea and Eukarya show mosaicism and a chimeric origin of genome repertoire. Viruses participates in gene transference mechanisms and also it is possible that they were involved in the origin of Eukaryotic cells (Bell 2009; Durzyńska and Goździcka-Józefiak 2015). Human genome shows in its structure

the viral, archaeal and bacterial origin (Tristem 2000; Georgiades and Raoult 2011). This reveals the rhizome-like processes of evolution as was stated by many authors (Raoult 2010; Merhej and Raoult 2012; Ramulu et al. 2012; McInerney et al. 2015; Levasseur et al. 2017).

There is no organism on Earth that has evolved and emerged alone but by complex processes of integration. This is evidenced in cell types origin, symbiogenetic mechanisms and the intricate metabolisms observed in symbiotic organisms (Margulis and Chapman 1998; Gilbert et al. 2010; Vannier-Santos and Lenzi 2011; Aanen and Eggleton 2017; Lane 2017). According to this perspective, the human organism can be understood as a superorganism (Figure 1) (Singh et al. 2013; Salvucci 2016; van der Meulen et al. 2016). This association is also called holobiont and was stated by Lynn Margulis (Cavalier-Smith 1992). There are also others similar systemic perspectives like hologenome, autocenosis and demo-cenosis (Rosenberg and Zilber-Rosenberg 2011; Savinov 2012).

Human superorganism has evolved and suffered several epidemiological transitions that caused serious changes in the microbiome. Along the human being existence on Earth there were processes called epidemiological transitions. These are complex changes in patterns of health and disease. These transitions were accompanied by demographic, economic and sociologic changes. This implies gradual but significant changes in patterns of livelihood, living conditions, population size, density and daily life (McKeown 2009). In relation with microbiome, we can consider two main epidemiological transitions: the earliest transition dates to pre-history (Neolithic period, approximately 10,000 years ago) across geographically widely separated cultures. It was characterised by the transformation from hunter-gatherer societies to agrarian societies and domestication of animals. Microbiome was more diverse and there were species that are currently considered parasites. Human microbiome in this transition contained *Salmonella spp.*, *Helicobacter pylori*, helminths, orto and paramyxovirus, herpesvirus.

The second transition occurs in the early modern period, late 19th and 20th centuries in developed countries and it is characterised by a shift in patterns of disease and mortality from primarily infectious diseases to what have come to be called "chronic" diseases. Life in cities, sanitisation, antibiotics, and change in diets were accompanied by increase in life expectancy (McKeown 2009). Microbiome was severely reduced or depleted. *Prevotella, Catenibacterium, Succinivibrio* and *Treponema* are the bacterial taxa that were lost in the transition from rural-traditional to urban industrial life-styles (Tito

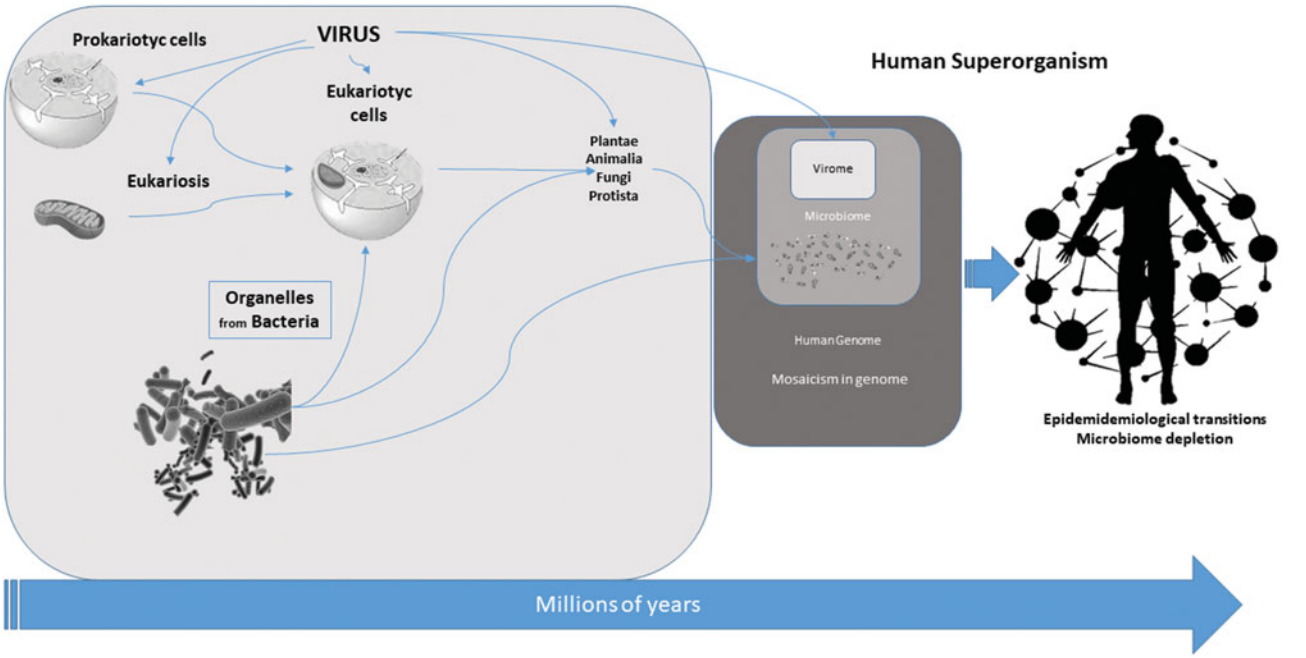


Figure 1. Scheme of integrative processes in evolution. Symbiogenesis is a main mechanism in evolution. The integration of symbionts resulted in new cell types and also new species. Virus have the ability to integrate into cells and to reorganise genomes. They are involved in emergence of different structures. Integration of host with its symbionts are observed in human superorganism. Microbiome constitute part of the human superorganism metabolism and virome is a genetic pool that respond to environmental changes.

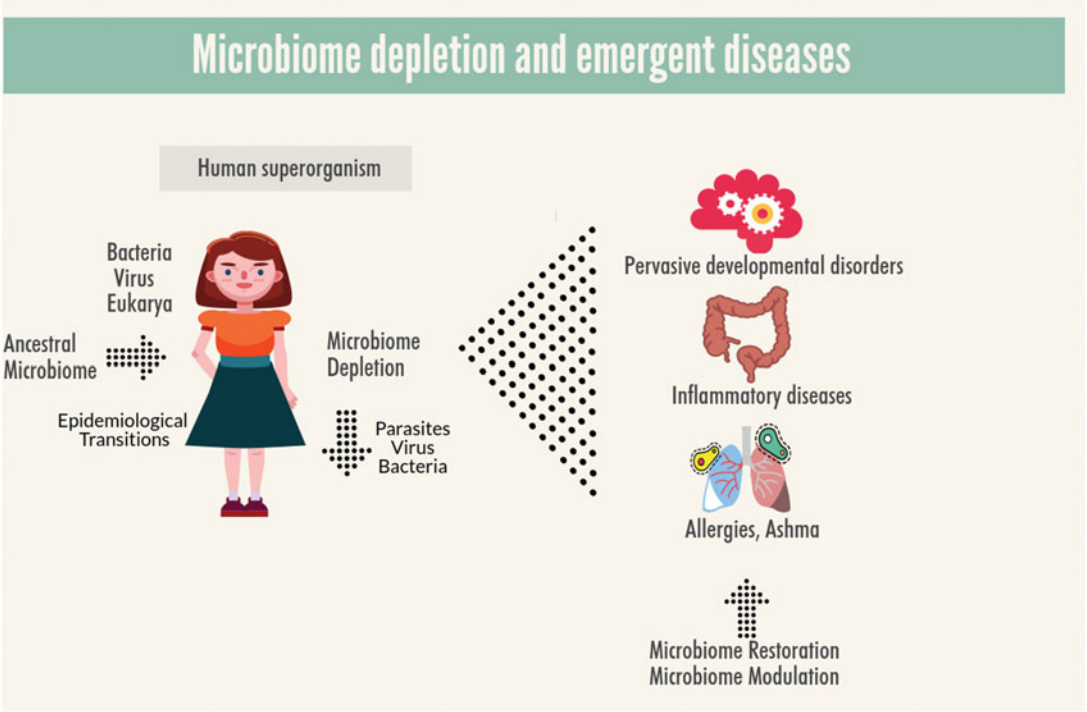


Figure 2. Human superorganism is a step of the integration processes observed along nature. During social and technological changes (epidemiological transitions) part of our microbiome was lost. Microbiome depletion causes the emergence of many diseases. Microbiome modulation by diet could restore functionality of the superorganism.

et al. 2012; Yatsunenko et al. 2012; Obregon-Tito et al. 2015; Schnorr et al. 2016). Ancestral diets and way of life have shown a more diverse microbiome and less inflammatory stress (Schnorr et al., 2014; Smits et al. 2017). Also, some microorganisms that are associated to illnesses in developed countries are part of normal microbiome in other regions (Marini et al. 2007).

In this last transition, the equilibrium of the whole system was altered and manifestations at immune level and western diseases emerged (Figure 2). Immunological, endocrine, metabolic and neurological alterations have arisen. Many human diseases like IBD, irritable bowel syndrome (IBS), Crohn disease, cancer, diabetes and asthma, allergies, obesity and metabolic syndrome are related to a change in this superorganism equilibrium. (Rook 2009; Foster and McVey Neufeld 2013; Arrieta et al. 2014; Paun and Danska 2015; Hayley et al. 2016; van der Meulen et al. 2016).

According to other authors, microbiome is also considered a "new organ" because of its relevance in health or disease of the human superorganism (Baquero and Nombela 2012; Evans et al. 2013; Clarke et al. 2014). This is another form to understand the superorganism: microbiome has a huge metabolic capacity to produce and regulate multiple compounds that modulate local metabolism. But these compounds also reach the circulation and influence the function of distal organs and systems. It is logical that its loss (microbiome depletion along evolution) cause significant effect on health/disease status like a "real" organ malfunction.

The ancestral diet based on high fibre intake maintains a well-balanced microbiome. The microbiota is structurally and functionally more diverse (with enhanced polysaccharide breakdown capacities, in particular) in rural and remote populations from developing countries than in urban industrialised populations (Schnorr et al. 2014; West et al. 2015; Turroni et al. 2016; Smits et al. 2017). The microbiota from traditional agrarian groups in Malawi, Venezuela and Burkina Faso were more diverse and similar to one another also in short chain fatty acids (SCFAs) production than to their Western counterparts. Western populations have depleted alpha diversity from birth through childbearing years and are missing bacterial taxa present in non-western populations with ancestral diet. Some bacterial taxa and functions may have disappeared from the gut during westernisation (Schnorr et al. 2014). *Prevotella* is overrepresented in ancestral diet and is associated with the increased capacity for complex carbohydrate metabolism. Other enriched genes are those coding for phosphoenolpyruvate carboxykinase (PEPCK) and intermediary enzymes in the pentose phosphate pathway (PPP).

The fermentation of pyruvate through these enzymes gives rise to intermediates like oxaloacetic acid or alanine, which can result in the production of SCFA or aromatic amino acids (Rampelli et al. 2015). An ancestral microbiome presents high capacity of the microbiota to accept many types of monosaccharides, enrichment in histidine metabolism, important pathways toward signal production and carbon recycling. It was also found a complementary functional structure of the microbiome to degrade and recycle complex macromolecules, most likely from the diet, into other substrates that are essential to the superorganism (Rampelli et al. 2015).

Migration is another important factor that alters microbiome structure. It has been reported that migration from non-western to western countries reduces the microbiome diversity. Residency in the United States increases the risk of obesity and other chronic diseases among immigrants relative to individuals of the same ethnicity that continue to reside in their country of birth (Vangay et al. 2018). It is related to pronounced increases (until 10-fold), in the ratio of *Bacteroides* to *Prevotella* and this loss of *Prevotella* strains was associated with decrease of carbohydrate-active enzymes dominant in the gut microbiota, including a near-complete loss of certain β-glucanases and other glycoside hydrolases that break down specific dietary fibres (Vangay et al. 2018).

The modernisation and westernisation was accompanied with increased ingest of high fat diet and low microbiome accessible carbohydrates (MACs) (carbohydrates that can be metabolically used by microbiome), that include dietary fibre (Sonnenburg and Sonnenburg 2014). Concomitantly, a progressive loss of diversity occurs. This damage is not recoverable upon the reintroduction of dietary fibre. Restoring the microbiome to its original state requires the administration of missing taxa in combination with dietary fibre consumption (Sonnenburg et al. 2016).

The negative effects of the depleted microbiome along epidemiological transitions, the impact of migration to western countries, the metabolic richness of microbiome observed in populations with ancestral diets (increased butyrate production and Bacteroides:Firmicutes ratio, higher *Prevotella* representation, wider metabolic functions) remarks the central importance of biodiversity of microbiome as a hallmark of health status (MacFabe et al. 2007; Schnorr et al. 2014; Sonnenburg and Sonnenburg 2014).

Summarising, the superorganismic approach has some major points: (1) human being is a superorganism, evolutionary built as an integration of eukaryotic host, microbiome and virome; (2) microbiome is a constitutive part of the human being and the evolutionary changes imply changes of the whole system; (3) during epidemiological transitions, microbiome was depleted and the system re-adapted to new equilibrium; (4) microbiome depletion is the loss of part of microorganism that contribute to the metabolic equilibrium and dysbiosis implies alteration of ecological structure of microbiome or less microbial richness; (5) the imbalance results in the emergence and increased incidence of several diseases related to immune and neuroendocrine alterations.

Modulation of microbiome

Gut microbiome imbalance is particularly associated with numerous inflammatory, immune and nervous system-related diseases (Salvucci 2016). The development of any of these diseases is directly and closely related to (and influenced by) the metabolism of the microbiome or diet, and the modulation of these factors can restore these disabilities. Many autoimmune and inflammatory diseases such as type 1 diabetes, arthritis, lupus, IBD and asthma can be treated by restoration or modulation of microbiome, i.e. re-establishing a more diverse and healthy microbiome. For instance, inflammatory diseases or allergies have shown to be alleviated by helminths (Hunter and McKay 2004; Reddy and Fried 2009; Weinstock and Elliott 2013).

Modulation of microbiome is intended to intervene in the symptomatology or causes of diseases in order to improve health status. This can be achieved by the action of certain bacteria that cause some beneficial effect (probiotics), by components of the diet (prebiotics) that modulate bacterial populations obtaining an integral improvement or by a combination of both (symbiotics) (Broekaert et al. 2011; Al-Sheraji et al. 2013; Vandenplas et al. 2015). The prebiotic effect in gut can be assessed by decreasing intestinal pathogens, increasing production of metabolites related to the activity of beneficial bacteria, decreasing toxic metabolites and for the effect on the microbiome structure (Al-Sheraji et al. 2013). Probiotics and prebiotics administration, helminthic therapies and faecal microbiome transplants are different strategies to achieve a restoration of the microbiome (Aggarwal et al. 2013; Singh et al. 2013; Vandenplas et al. 2015) (Table 1).

The restoration of the microbiome has been studied mainly in metabolic (obesity, metabolic syndrome) or inflammatory diseases (IBS, Crohn, IBD) (Manichanh et al. 2010; Davis 2016). Duvallet et al. (2017) based

Table 1. Some successful reports of microbiome modulation to restore health.

Treatment	Disease type	Result	Reference
Prebiotics			
Inulin	Pouchitis	20 patients. Increased butyrate concentrations, lowered pH, reduction of inflammation of the mucosa of the ileal reservoir.	(Welters et al. 2002)
FOS/GOS	Eczema, asthma	Four studies enrolling 1428 infants. Individual studies reported a significant reduction in asthma and eczema from supplementation with a mixture of galacto- and fructo-oligosaccharide.	(Osborn and Sinn 2013)
GOS	Autism	41 autistic children. Children on exclusion diets reported significantly lower scores of abdominal pain and bowel movement. We observed improvements in anti-social behaviour.	(Grimaldi et al. 2018)
FOS	Overweight or obesity	44 participants. Children who consumed OI had significant decreases in body weight z-score, percent body fat and percent trunk fat	(Nicolucci et al. 2017)
Probiotics			
Lactobacillus rhamnosus GG; *L. rhamnosus* LC705; *Bifidobacterium breve* Bb99; *Propionibacterium freudenreichii ssp. shermanii* JS	Eczema	1223 pregnant women. Probiotic treatment showed no effect on the incidence of all allergic diseases by age 2 years but significantly prevented eczema and, especially, atopic eczema.	(Kukkonen et al. 2007)
VSL#3	Ulcerative Colitis	32 patients. Combined induction of remission/response rate of 77% with no adverse events.	(Bibiloni et al. 2005)
Lactobacillus rhamnosus GG, *L. acidophilus* La-5 and *Bifidobacterium animalis subsp. lactis* Bb-12	Allergies	415 pregnant women. At 2 years, 138 and 140 children in the probiotic and the placebo groups, respectively, were assessed. Probiotics given to nonselected mothers reduced the cumulative incidence of atopic dermatitis, but had no effect on atopic sensitisation.	(Dotterud et al. 2010)
Bifidobacterium longum	IBD	120 patients. Total Inflammatory Bowel Disease Questionnaires (IBDQ) scores improved within groups by the end of the trial (probiotics 162 to 169)	(Fujimori et al. 2009)
Synbiotic			
Bifidobacterium longum and psyllium.	IBD	32 patients with UC on synbiotic therapy experienced greater quality-of-life changes than patients on probiotic or prebiotic treatment.	(Fujimori et al. 2009)
Helminthic therapies			
Necator	Crohn	9 patients. After 20 weeks, the IBD questionnaire was improved (mean 151 (14) vs. 179 (20)) and the four week accumulated Crohn's disease activity index (CDAI) scores was decreased.	(Croese et al. 2006)
Necator americanus	Asthma	32 individuals with asthma. The treatment was well tolerated but did not result in significant improvement in bronchial responsiveness or other measures of asthma.	(Feary et al. 2010)
Trichuris suis	Crohn	29 patients. 23 patients (79.3%) responded and 21 (72.4%) remitted.	(Summers et al. 2005a, 2005b)
Trichuris suis	Ulcerative colitis	54 patients. Improvement occurred in 43.3% of patients.	(Summers et al. 2005a, 2005b)
Trichuris suis	Allergic rhinitis	100 patients. The treatment induced a substantial clinical (diarrhoea) and immunologic response as evidence of infection, but had no therapeutic effect on allergic rhinitis.	(Bager et al. 2010)
Necator americanus	Coeliac disease	20 patients. Wheat challenge after 20 weeks. Doses of 5–10 larvae. The treatment was well tolerated and immunological analysis demonstrated reduced inflammatory cytokine (IFN-γ and IL-17) responses.	(McSorley et al. 2011)
Faecal microbiome transplant			
FMT	Crohn	1 patient with Crohn's colitis with clinical remission and endoscopic improvement after a single faecal microbiota transplantation.	(Bak et al. 2017)
FMT	Crohn	1 patient with Crohn's colitis, previously failing an immunosuppressant, who achieved clinical, endoscopic, and histologic remission	(Kao et al. 2014)

(*continued*)

Table 1. Continued.

Treatment	Disease type	Result	Reference
		after a single faecal microbiota transplantation.	
FMT	CDI	4 reports with 89–96% of clinical success	(Gianotti and Moss 2017)
FMT	CDI	1 patient. A successful FMT in a patient with severe therapy-refractory CDI.	(Schneider et al. 2018)

VSL#3: *Bifidobacterium breve*, *Bifidobacterium longum*, *Bifidobacterium infantis*, *Lactobacillus acidophilus*, *Lactobacillus plantarum*, *Lactobacillus paracasei*, *Lactobacillus bulgaricus* and *Streptococcus thermophilus*.
CDI: *Clostridium difficile* infection.

in 28 studies of dysbiosis associated diseases, found different patterns of microbiome alterations that differ in their directionality. Dysbiosis can be associated to a depletion of health-associated bacteria, the increase of some pathogenic bacteria or it can be characterised by a broad restructuring of gut microbial communities. There are diseases with enriched-associated microbes, like colorectal cancer, which could be targeted with narrow-spectrum anti-microbials. There are other diseases, like IBD, that present depleted genera. In these cases, probiotics that replace missing taxa or prebiotic intervention may be better strategies than antimicrobials (Duvallet et al. 2017).

In obesity and metabolic syndrome, there are several associated features: increased intestinal permeability, LPS translocation and systemic low-grade inflammation, resulting in a decreased glucose tolerance and hyperphagic behaviour. The disruption of the normal intestinal barrier enhances the access of Gram-negative cell wall parts such as lipopolysaccharide (LPS) endotoxin into blood. An increase in blood LPS levels (0.5–1.0 ng/kg) can provoke an inflammatory immune response. It can also interfere with cognition and cause emotional symptoms (Sandiego et al. 2015). The use of probiotics is trying to "rebalance" normal commensal flora by the deliberate introduction of beneficial microorganisms. It has shown several clinical success in animal models (Bercik et al. 2011; Saad et al. 2013; Savignac et al. 2013). *Bifidobacterium* strains have shown effective activity to restore some manifestations, contributing to re-establish the levels of cytokines and hormones like leptin involved in the regulation of appetite (Cano et al. 2013; Agusti et al. 2018; Bagarolli et al. 2017).

Faecalibacterium prausnitzii is a major member of the Firmicutes phylum and one of the most abundant bacteria in the healthy human microbiota. *F. prausnitzii* depletion has been reported in several intestinal disorders, IBS, colorectal cancer, obesity, coeliac disease and more consistently in Crohn's disease patients and it became a crucial strain to be considered in modulation of microbiome. Recent studies confirm that it produces SCFA and the butyrate production has been correlated to the capacity to induce IL-10, an anti-inflammatory cytokine, that can postulate this specie as a next-generation probiotic (Martín et al. 2017).

Regarding the mycobiome, it is known that an increase in diversity is associated with the progression of diseases like IBD (Crohn's disease and ulcerative colitis), but it remains unknown the role of specific phyla in the disease and if this richness is the cause or the consequence (Cui et al. 2013). Some strains of the mycobiome may also play a beneficial role. The promotion of its growth could result in an improvement of the disease. For instance, *Saccharomyces boulardii* has shown to be effective for the treatment of diarrhoeal diseases (Palma et al. 2015) and IBD (Thomas et al. 2011). The increase in gut permeability is observed in inflammatory diseases and obesity as a consequence of a reduction in tight junction protein expression. This can be reverted after probiotic treatment or microbiome modulation. SCFA are the molecules involved in this reversion (Rios-Covian et al. 2013). Yeasts and fungi, such as *S. cerevisiae* and *Candida* species and *Cryptococcus neoformans* are some strains able to produce SCFA. They represent potential targets to take into account considering their contribution to alleviate the process (Enaud et al. 2018). It opens the possibility to search beneficial fungal strains that can be modulate by diet.

Microbiome and neurodevelopmental disorders

Can we move forward in microbiome based treatment according to the superorganismic approach? This model can be extended to complex illnesses in which the genetic background is determinant. In these cases, the modulation of certain factors can alter the course of an illness because the genetic alteration can be compensated by the action of other genes. In the same way, a metabolic change could be restored by modulation of microbiome that influences the systemic metabolism.

Microbiome–brain axis is the two-way communication pathway between the central nervous system and the intestine. The metabolism of microorganisms acts not only locally, but also generates many changes at the level of neurotransmitters or hormones (Allen et al. 2013; O'Mahony et al. 2015). It plays a decisive role in the metabolism of GABA and serotonin which are related to anxiety, depression and mood states (Bercik et al. 2010; Foster and McVey Neufeld 2013). Moreover, it was remarked that gut microbiome influences brain chemistry and behaviour independently from the autonomic nervous system (Bercik et al. 2012; Foster and McVey Neufeld 2013).

The reciprocal influence of microbiome and brain is observed in inflammatory diseases. IBD are often accompanied by dysbiosis and mental disorders such as irritability, anxiety and depression (Lee and Chua 2011; Hayley et al. 2016). Depression and anxiety are associated with a more intense disease course and dysbiosis can contribute to psychiatric disorders development in patients with intestinal symptoms (2010; Collins et al. 2012; Borre et al. 2014).

The crucial point in the increase of incidence in such illnesses is that there is a missing symbiotic counterpart that built the human being. It was lost simultaneously with social changes that included the use and overuse of antibiotics, sanitisation and western diets. The effect of that loss is also observed at brain level. Neurodevelopmental disorders present concomitant gut alterations (Frye et al. 2016) and they can be now studied focussing the research under the human superorganism approach since the whole organism presents neuroendocrine alterations associated with microbiome dysbiosis (Rosenfeld 2015).

Pervasive developmental disorders (PDDs) are among the disorders related to altered microbiome (Rosenfeld 2015). PDD includes autism spectrum disorders (ASDs) and Rett syndrome (RTT). ASD affects a significant number of individuals worldwide and the prevalence continues to grow. Gastrointestinal (GI) tract symptoms are very common in individuals with ASD, many times co-occurring with abnormalities in mitochondrial function. Concomitantly, it is observed a reduced number of bifidobacteria and increased *Clostridium* spp., *Desulfovibrio* spp., *Sutterella* spp. and Veillonellaceae (Li et al. 2017). Moreover, there is an increase in the Firmicutes:Bacteroidetes ratio in autistic subjects due to a reduction of the Bacteroidetes relative abundance. Strati et al. (2017) reported a decrease in the relative abundance of *Alistipes*, *Bilophila*, *Dialister* and *Parabacteroides*. Also, the relative abundance of the fungal genus *Candida* was more than double in the autistic than neurotypical subjects (Strati, et al. 2017).

Diet influences the microbiome and prebiotics acquire a fundamental role in health and disease status (Hattori and Taylor 2009; Flint 2012; Martinez et al. 2013). A mechanism of action is the production of SCFA. Butyrate is largely produced by different strains of the microbiome and it has shown profound central nervous system effects on mood and behaviour in mice (Schroeder et al. 2007). SCFA regulate the transcription of genes potentially involved in immune and metabolic diseases. The mechanisms involve G protein-coupled protein receptors – GPR43, GPR41 and GPR109A (also known as: FFAR2, FFAR3 and NIACR1, respectively) – and histone deacetylases (Kasubuchi et al. 2015; Corrêa-Oliveira et al. 2016).

Microbiome and SCFA levels are critical environmental triggers for ASD. They have broad effects on cellular systems and have impact on physiology and behaviour. Propionate can induce ASD-like behaviour in animal models (Foley et al. 2014). In human ASD patients have elevated faecal propionate levels. Moreover, propionate is involved in gap junction dysfunction, mitochondrial disorder and impairments in B12, glutathione or carnitine metabolism (MacFabe 2012; Frye et al. 2016).

Probiotics have shown successful anti-inflammatory, anti-anxiety and anti-depressive effect in human and animal studies (Paun and Danska 2015). Probiotic intervention was found to be associated with higher levels of selected SCFA (Hemalatha et al. 2017) and there is increasing evidence of successful use of probiotics in diseases related to the nervous system such as ASDs or schizophrenia (Hsiao et al. 2013; de Theije et al. 2014; Severance et al. 2016). Synbiotic combinations of *Lactobacillus acidophilus* NCFM and *Bifidobacterium animalis subsp. lactis* Bl-04 with prebiotics (melibiose, xylobiose, raffinose and maltotriose) significantly increasing SCFA levels, especially acetic and butyric acid (van Zanten et al. 2012). Recently, it was reported that exclusion diet (gluten free and casein free diets) plus treatment with galactooligosacharides significantly improves behavioural traits in ASD children. Also, general trend of reduction in GI problems was observed. Bacterial populations positively associated with GOS supplementation in ASD children were *Bifidobacterium spp., Ruminococcus spp.,* Lachnospiraceae family, *Eubacterium dolchum* and Mogibacteriaceae. Additionally, supplementation increased diversity in microbiome composition (Grimaldi et al. 2018). The search of strains or synbiotics combinations that can restore the altered levels of

different SCFA associated with diseases is a therapeutic approach to relieve symptoms or change the course of the disorder.

RTT is a progressive neurodevelopmental disorder associated with intellectual disability that is caused by loss of function mutations in the transcriptional regulator methyl CpG-binding protein 2 (MECP2). These alterations result, among other effects, in altered levels of BDNF (brain-derived neurotrophic factor) (Katz 2014; Xu et al. 2014; Theoharides et al. 2015). It is observed that modulation of microbiome through administration of probiotics can restore some levels of neurotransmitters that can alleviate some neurological manifestations (Bercik et al. 2011; Foster and McVey Neufeld 2013). Bacterial strains like *Bifidobacterium longum* (Bercik et al. 2010, 2011), have the ability to influence metabolism of neurotransmitters and neurotrophic factors, and could represent a target of treatment of these disabilities. The administration of prebiotic FOS has shown to upregulate the BDNF expression and this effect can be mediated by Bifidobacteria (Savignac et al. 2013).

Multiple lines of evidence suggest the direct and indirect intervention of BDNF in autism disorders, playing a key role in brain development and plasticity (Chapleau et al. 2009; Ricci et al. 2013). BDNF has a trophic effect on specific neuronal populations during brain development and maturation, decisively influencing survival, morphology, differentiation and neuronal synapses (Correia et al. 2010; Samaco and Neul 2011; Allen et al. 2013). Little is known about the molecular mechanisms by which mutations in MeCP2 cause neurodevelopmental defects leading to the complex clinical course of RTT. It has been observed that BDNF expression is decreased and pharmacological treatment that could reverse some manifestations of RTT is associated with increased expression of BDNF (Chang et al. 2006; Chapleau et al. 2009; Larimore et al. 2009). The regulation of this neurotrophin becomes a fundamental target for the modulation of the brain anatomy in cases of autistic spectrum syndrome and PDD, as RTT. In these cases, the synaptic activity and neuronal development are compromised (Halepoto et al. 2014). Current studies on RTT and PDD have not focussed, however, on the epigenetic action of the microbiome and its demonstrated influence on health and disease status and the intrinsic ability to intervene in the gut-brain axis (Cryan and Dinan 2012; Foster and McVey Neufeld 2013; Borre et al. 2014). In RTT and ASD patients, it is observed an increase in *Candida* abundance (Strati et al. 2017). The effectiveness of any probiotic or synbiotic treatment should take into account this factor since its importance at local and systemic level (Enaud et al. 2018).

Beyond the mechanisms that underlie the action of microorganisms or certain species or groups (which may be a subject of future research) at first it is interesting to study the variability of the microbiome in RTT and to intervene through a bifidogenic prebiotic diet to treat the symptoms resulting from altered levels of BDNF. An increase in *Bifidobacterium* spp. is a desirable effect because it is a depleted group in microbiome of ASD patients. Moreover, *Bifidobacterium* strains exert beneficial effect at brain level. Some *Bifidobacterium* strains can intervene in the BDNF regulation (O'Sullivan et al. 2011), for instance, *B. longum* normalises anxiety-like behaviour and levels of BDNF in hippocampus in mice with colitis. *B. infantis* attenuates stress-induced immunological changes in children and depressive-like behaviour in adults (Savignac et al. 2014). Moreover, it was observed that *Bifidobacterium* can restore the hippocampal BDNF and c-Fos expression (Lee and Chua 2011). Other strains can influence neurological symptoms. *Bacteroides fragilis* can improve intestinal permeability and defects in anxiety, communication and sensory motor behaviour. It modifies the levels of several metabolites increased in autism models (Hsiao et al. 2013).

It becomes interesting to develop synbiotic formulas with one of these selected strains and the best prebiotic that promotes its growth and persistence in the gut to modulate brain function. Mice without a functional MeCP2 gene (KO mice) constitute a murine model for RTT. These mice showed defects in the postnatal refinement of this circuit in response to synaptic activity (exposure to odorants). Lack of refinement was associated with deficits in BDNF production and changes in the expression of guiding molecules that regulate axonal convergence. This (genetic) deficiency of BDNF can be reversed with prolonged treatment with ampakines, allosteric modulators that interact with glutamate receptor, despite the underlying mutation (Degano et al. 2014). Different studies agree on the likely pathophysiological role of BDNF levels, deregulated by mutations or alterations in the MECP2 regulator.

The microbiome actively participates in the regulation of neurotransmitters and neurotrophins such as BDNF. There are many promising prebiotics that are widely studied. Arabinoxylans (AX) are interesting candidates to promote the growth of beneficial strains that could relieve symptoms of different

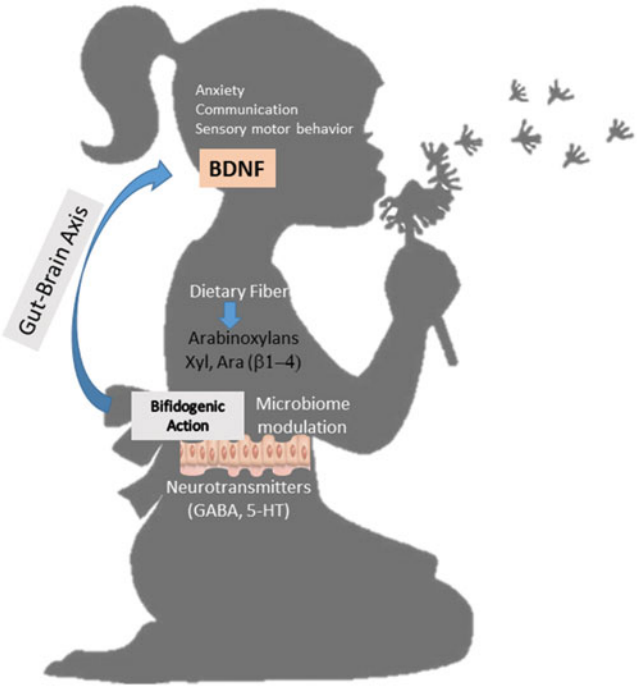


Figure 3. Microbiome-diet treatments in PDD. Components of diets (prebiotics, arabinoxylans) can modulate microbiome landscape, bacterial ratios and metabolism. The proliferation of *Bifidobacterium* strains can modulate the levels of BDNF and the process can influence and alleviate symptoms of PDD. The beneficial changes are observed in both gastrointestinal and neurological levels.

neurodevelopmental diseases (Broekaert et al. 2011; Rivière et al. 2014). Sanchez et al. (2009) showed that AX are able to modulate the microbial metabolism in the distal colon (Sanchez et al. 2009). Truchado et al. (2017) have shown that AX extracts significantly increased the relative abundance of *Bifidobacterium* in a gut model (Truchado et al. 2017). *B. longum* and *B. adolescentis* grow in the presence of AX using molecules provided by the arabinose degradation by bacteria that produce xylanase (Pastell et al. 2009).

The ability of certain *Bifidobacterium* species to intervene in the so-called gut-brain axis through the regulation of certain neurotransmitters and neurotrophic factors (mainly, BDNF) opens the opportunity of evaluating the effect of AX in models of PDD, such as RTT (Figure 3). A synbiotic including BDNF-regulator probiotic strains and AX is a strategy to explore the possibility to alleviate gut symptoms in neurodevelopmental diseases.

Conclusions

Superorganismic approach becomes relevant because it consists in a theoretic framework to understand some physiological manifestations that co-occur with mental disorders as an imbalance at immunological or ecological level. Microbiome depletion occurred along epidemiological transitions which involved social and technological changes.

GI tract dysbiosis accompanied metabolic alterations and systemic effects. The modulation or restoration of microbiome has shown positive results in different autoimmune or inflammatory diseases like Crohn or IBS.

The modulation of microbiome (bacteria or fungi) can be a strategic therapy to help alleviate some manifestations of neurodevelopmental disorders. The influence of certain *Bifidobacterium* strains in the balance of neurotrophins opens the possibility of using prebiotic (like arabinoxylans) or symbiotic (prebiotics + bacteria/fungi probiotics) in order to ameliorate the gut or neurological symptoms in RTT.

Acknowledgements

The author would like to thank Prof. Brenda Burgos for their assistance and corrections to improve English use.

Disclosure statement

The author has no interests which might be perceived as posing a conflict or bias.

Funding

The author would like to thank CONICET and ANPYCT-FONCYT (grant PICT 2015-0606) for the financial support.

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References

Aanen DK, Eggleton P. 2017. Symbiogenesis: beyond the endosymbiosis theory? J Theor Biol. 434:99–103.

Aggarwal J, Swami G, Kumar M. 2013. Probiotics and their effects on metabolic diseases: an update. *J Clin Diagn Res*. 7:173–177.

Agusti A, Moya-Pérez A, Campillo I, Montserrat-de la Paz S, Cerrudo V, Perez-Villalba A, Sanz Y. 2018. *Bifidobacterium pseudocatenulatum* CECT 7765 ameliorates neuroendocrine alterations associated with an exaggerated stress response and anhedonia in obese mice. *Mol Neurobiol*. 55:5337–5352.

Allen SJ, Watson JJ, Shoemark DK, Barua NU, Patel NK. 2013. GDNF, NGF and BDNF as therapeutic options for neurodegeneration. Pharmacol Ther. 138:155–175.

Al-Sheraji SH, Ismail A, Manap MY, Mustafa S, Yusof RM, Hassan FA. 2013. Prebiotics as functional foods: a review. J Funct Foods. 5:1542–1553.

Arrieta M-C, Stiemsma LT, Amenyogbe N, Brown EM, Finlay B. 2014. The intestinal microbiome in early life: health and disease. *Front Immunol*. 5:427.

Bagarolli RA, Tobar N, Oliveira AG, Araújo TG, Carvalho BM, Rocha GZ, Vecina JF, Calisto K, Guadagnini D, Prada PO, et al. 2017. Probiotics modulate gut microbiota and improve insulin sensitivity in DIO mice. *J Nutr Biochem*. 50:16–25.

Bager P, Arnved J, Rønborg S, Wohlfahrt J, Poulsen LK, Westergaard T, Petersen HW, Kristensen B, Thamsborg S, Roepstorff A, et al. 2010. *Trichuris suis* ova therapy for allergic rhinitis: a randomized, double-blind, placebo-controlled clinical trial. *J Allergy Clin Immunol*. 125: 123–130.e1–3.

Bak SH, Choi HH, Lee J, Kim MH, Lee YH, Kim JS, Cho Y-S. 2017. Fecal microbiota transplantation for refractory Crohn's disease. Intest Res. 15:244–248.

Bakhtiar SM, LeBlanc JG, Salvucci E, Ali A, Martin R, Langella P, Chatel JM, Miyoshi A, Bermúdez-Humarán LG, Azevedo V. 2013. Implications of the human microbiome in inflammatory bowel diseases. *FEMS Microbiol Lett*. 342:10–17.

Baquero F, Nombela C. 2012. The microbiome as a human organ. Clin Microbiol Infect. 18(Suppl. 4):2–4.

Bell PJL. 2009. The viral eukaryogenesis hypothesis: a key role for viruses in the emergence of eukaryotes from a prokaryotic world environment. Ann NY Acad Sci. 1178: 91–105.

Benson AK. 2016. The gut microbiome—an emerging complex trait. Nat Genet. 48:1301–1302.

Bercik P, Collins SM, Verdu EF. 2012. Microbes and the gut-brain axis. Neurogastroenterol Motil. 24:405–413.

Bercik P, Park AJ, Sinclair D, Khoshdel A, Lu J, Huang X, Deng Y, Blennerhassett PA, Fahnestock M, Moine D, et al.etal. 2011. The anxiolytic effect of *Bifidobacterium longum* NCC3001 involves vagal pathways for gut–brain communication. *Neurogastroenterol Motil*. 23:1132–1139.

Bercik P, Verdu EF, Foster JA, Macri J, Potter M, Huang X, Malinowski P, Jackson W, Blennerhassett P, Neufeld KA, et al. 2010. Chronic gastrointestinal inflammation induces anxiety-like behavior and alters central nervous system biochemistry in mice. *Gastroenterology*. 139:2102–2112.e1.

Bibiloni R, Fedorak RN, Tannock GW, Madsen KL, Gionchetti P, Campieri M, De Simone C, Sartor RB. 2005. VSL#3 probiotic-mixture induces remission in patients with active ulcerative colitis. Am J Gastroenterol. 100:1539–1546.

Borre YE, O'Keeffe GW, Clarke G, Stanton C, Dinan TG, Cryan JF. 2014. Microbiota and neurodevelopmental windows: implications for brain disorders. Trends Mol Med. 20:509–518.

Broekaert WF, Courtin CM, Verbeke K, Van de Wiele T, Verstraete W, Delcour JA. 2011. Prebiotic and other health-related effects of cereal-derived arabinoxylans, arabinoxylan-oligosaccharides, and xylooligosaccharides. Crit Rev Food Sci Nutr. 51:178–194.

Brucker RM, Bordenstein SR. 2012. Speciation by symbiosis. Trends Ecol Evol (Amst). 27:443–451.

Cano PG, Santacruz A, Trejo FM, Sanz Y. 2013. Bifidobacterium CECT 7765 improves metabolic and immunological alterations associated with obesity in high-fat diet-fed mice. Obesity (Silver Spring MD). 21: 2310–2321.

Cavalier-Smith T. 1992. Symbiosis as a source of evolutionary innovation: speciation and morphogenesis: edited by Lynn Margulis and René Fester, MIT Press, 1991. £33.75 hbk (xiii + 454 pages) ISBN 0 262 13269 9. Trends Ecol Evol. 7:422–423.

Chang Q, Khare G, Dani V, Nelson S, Jaenisch R. 2006. The disease progression of Mecp2 mutant mice is affected by the level of BDNF expression. Neuron. 49:341–348.

Chapleau CA, Larimore JL, Theibert A, Pozzo-Miller L. 2009. Modulation of dendritic spine development and plasticity by BDNF and vesicular trafficking: fundamental roles in neurodevelopmental disorders associated with mental retardation and autism. J Neurodev Disord. 1: 185–196.

Chuong EB, Elde NC, Feschotte C. 2017. Regulatory activities of transposable elements: from conflicts to benefits. Nat Rev Genet. 18:71–86.

Clarke G, Stilling RM, Kennedy PJ, Stanton C, Cryan JF, Dinan TG. 2014. Minireview: gut microbiota: the neglected endocrine organ. Mol Endocrinol (Baltim MD). 28:1221–1238.

Collins SM, Surette M, Bercik P. 2012. The interplay between the intestinal microbiota and the brain. Nat Rev Microbiol. 10:735–742.

Colson P, Levasseur A, La Scola B, Sharma V, Nasir A, Pontarotti P, Caetano-Anollés G, Raoult D. 2018. Ancestrality and mosaicism of giant viruses supporting the definition of the fourth TRUC of microbes. Front Microbiol. 9:2668.

Corrêa-Oliveira R, Fachi JL, Vieira A, Sato FT, Vinolo MAR. 2016. Regulation of immune cell function by short-chain fatty acids. Clin Transl Immunol. 5:e73.

Correia CT, Coutinho AM, Sequeira AF, Sousa IG, Lourenço Venda L, Almeida JP, Abreu RL, Lobo C, Miguel TS, Conroy J, et al. 2010. Increased BDNF levels and NTRK2 gene association suggest a disruption of BDNF/TrkB signaling in autism. *Genes Brain Behav*. 9: 841–848.

Croese J, O'Neil J, Masson J, Cooke S, Melrose W, Pritchard D, Speare R. 2006. A proof of concept study establishing *Necator americanus* in Crohn's patients and reservoir donors. Gut. 55:136–137.

Cryan JF, Dinan TG. 2012. Mind-altering microorganisms: the impact of the gut microbiota on brain and behaviour. Nat Rev Neurosci. 13:701–712.

Cui L, Morris A, Ghedin E. 2013. The human mycobiome in health and disease. Genome Med. 5:63.

Davis CD. 2016. The gut microbiome and its role in obesity. Nutr Today. 51:167–174.

de Theije CGM, Wopereis H, Ramadan M, van Eijndthoven T, Lambert J, Knol J, Garssen J, Kraneveld AD, Oozeer R. 2014. Altered gut microbiota and activity in a murine model of autism spectrum disorders. *Brain Behav Immun*. 37:197–206.

Degano AL, Park MJ, Penati J, Li Q, Ronnett GV. 2014. MeCP2 is required for activity-dependent refinement of olfactory circuits. Mol Cell Neurosci. 59:63–75.

Dotterud CK, Storrø O, Johnsen R, Øien T. 2010. Probiotics in pregnant women to prevent allergic disease: a randomized, double-blind trial. Br J Dermatol. 163: 616–623.

Durzyńska J, Goździcka-Józefiak A. 2015. Viruses and cells intertwined since the dawn of evolution. Virol J. 12:169.

Duvallet C, Gibbons SM, Gurry T, Irizarry RA, Alm EJ. 2017. Meta-analysis of gut microbiome studies identifies disease-specific and shared responses. Nat Commun. 8: 1784.

Enaud R, Vandenborght L-E, Coron N, Bazin T, Prevel R, Schaeverbeke T, Berger P, Fayon M, Lamireau T, Delhaes L. 2018. The mycobiome: a neglected component in the microbiota-gut-brain axis. *Microorganisms*. 6:1.

Evans JM, Morris LS, Marchesi J. 2013. The gut microbiome: the role of a virtual organ in the endocrinology of the host. *J Endocrinol*.218:R37–R47.

Feary JR, Venn AJ, Mortimer K, Brown AP, Hooi D, Falcone FH, Pritchard DI, Britton JR. 2010. Experimental hookworm infection: a randomized placebo-controlled trial in asthma. Clin Exp Allergy. 40:299–306.

Feschotte C. 2008. Transposable elements and the evolution of regulatory networks. Nat Rev Genet. 9:397–405.

Flint HJ. 2012. The impact of nutrition on the human microbiome. Nutr Rev. 70:S10–S13.

Foley KA, Ossenkopp K-P, Kavaliers M, MacFabe DF. 2014. Pre- and neonatal exposure to lipopolysaccharide or the enteric metabolite, propionic acid, alters development and behavior in adolescent rats in a sexually dimorphic manner. *PLoS One*. 9:e87072.

Foster JA, McVey Neufeld K-A. 2013. Gut-brain axis: how the microbiome influences anxiety and depression. Trends Neurosci. 36:305–312.

Frye RE, Rose S, Chacko J, Wynne R, Bennuri SC, Slattery JC, Tippett M, Delhey L, Melnyk S, Kahler SG, et al. 2016. Modulation of mitochondrial function by the microbiome metabolite propionic acid in autism and control cell lines. *Transl Psychiatry*. 6:e927.

Fugmann SD. 2010. The origins of the Rag genes-from transposition to V(D)J recombination. Semin Immunol. 22:10–16.

Fujimori S, Gudis K, Mitsui K, Seo T, Yonezawa M, Tanaka S, Tatsuguchi A, Sakamoto C. 2009. A randomized controlled trial on the efficacy of synbiotic versus probiotic or prebiotic treatment to improve the quality of life in patients with ulcerative colitis. Nutrition. 25:520–525.

Georgiades K, Raoult D. 2011. The rhizome of *Reclinomonas americana*, *Homo sapiens*, *Pediculus humanus* and *Saccharomyces cerevisiae* mitochondria. Biol Direct. 6:55.

Gianotti RJ, Moss AC. 2017. Fecal microbiota transplantation: from clostridium difficile to inflammatory bowel disease. Gastroenterol Hepatol (NY). 13:209–213.

Gilbert SF, McDonald E, Boyle N, Buttino N, Gyi L, Mai M, Prakash N, Robinson J. 2010. Symbiosis as a source of selectable epigenetic variation: taking the heat for the big guy. Philos Trans R Soc Lond B Biol Sci. 365: 671–678.

Grimaldi R, Gibson GR, Vulevic J, Giallourou N, Castro-Mejía JL, Hansen LH, Leigh Gibson E, Nielsen DS, Costabile A. 2018. A prebiotic intervention study in children with autism spectrum disorders (ASDs). *Microbiome*. 6:133.

Halepoto D, Bahir S, Al-Ayadhi LY. 2014. Possible role of brain-derived neurotrophic factor (BDNF) in autism spectrum disorder: current status. J Coll Phys Surg. 24: 274–278.

Hallen-Adams HE, Suhr MJ. 2017. Fungi in the healthy human gastrointestinal tract. Virulence. 8:352–358.

Hattori M, Taylor TD. 2009. The human intestinal microbiome: a new frontier of human biology. DNA Res. 16: 1–12.

Hayley S, Audet M-C, Anisman H. 2016. Inflammation and the microbiome: implications for depressive disorders. Curr Opin Pharmacol. 29:42–46.

Hemalatha R, Ouwehand AC, Saarinen MT, Prasad UV, Swetha K, Bhaskar V. 2017. Effect of probiotic supplementation on total lactobacilli, bifidobacteria and short chain fatty acids in 2–5-year-old children. *Microb Ecol Health Dis*. 28:1298340.

Hsiao EY, McBride SW, Hsien S, Sharon G, Hyde ER, McCue T, Codelli JA, Chow J, Reisman SE, Petrosino JF, et al. 2013. Microbiota modulate behavioral and physiological abnormalities associated with neurodevelopmental disorders. *Cell*. 155:1451–1463.

Hunter MM, McKay DM. 2004. Review article: helminths as therapeutic agents for inflammatory bowel disease. Aliment Pharmacol Ther. 19:167–177.

Iliev ID, Leonardi I. 2017. Fungal dysbiosis: immunity and interactions at mucosal barriers. Nat Rev Immunol. 17: 635–646.

Jablonka E, Raz G. 2009. Transgenerational epigenetic inheritance: prevalence, mechanisms, and implications for the study of heredity and evolution. Q Rev Biol. 84: 131–176.

Jones BV. 2010. The human gut mobile metagenome: a metazoan perspective. Gut Microbes. 1:415–431.
Kao D, Hotte N, Gillevet P, Madsen K. 2014. Fecal microbiota transplantation inducing remission in Crohn's colitis and the associated changes in fecal microbial profile. J Clin Gastroenterol. 48:625–628.
Kasubuchi M, Hasegawa S, Hiramatsu T, Ichimura A, Kimura I. 2015. Dietary gut microbial metabolites, short-chain fatty acids, and host metabolic regulation. Nutrients. 7:2839–2849.
Katz DM. 2014. Brain-derived neurotrophic factor and Rett syndrome. Handb Exp Pharmacol. 220:481–495.
Koleva PT, Bridgman SL, Kozyrskyj AL. 2015. The infant gut microbiome: evidence for obesity risk and dietary intervention. Nutrients. 7:2237–2260.
Kozo-Polyansky BM, Margulis L. 2010. Symbiogenesis: a new principle of evolution. Fet V, translator. Cambridge, MA: Paperbackshop UK Import.
Kramer P, Bressan P. 2015. Humans as superorganisms: how microbes, viruses, imprinted genes, and other selfish entities shape our behavior. Perspect Psychol Sci. 10: 464–481.
Kukkonen K, Savilahti E, Haahtela T, Juntunen-Backman K, Korpela R, Poussa T, Tuure T, Kuitunen M. 2007. Probiotics and prebiotic galacto-oligosaccharides in the prevention of allergic diseases: a randomized, double-blind, placebo-controlled trial. J Allergy Clin Immunol. 119:192–198.
Lane N. 2017. Serial endosymbiosis or singular event at the origin of eukaryotes? J Theor Biol. 434:58–67.
Larimore JL, Chapleau CA, Kudo S, Theibert A, Percy AK, Pozzo-Miller L. 2009. Bdnf overexpression in hippocampal neurons prevents dendritic atrophy caused by Rett-associated MECP2 mutations. Neurobiol Dis. 34:199–211.
Lee YY, Chua ASB. 2011. Influence of gut microbes on the brain-gut axis (Gut 2011;60:307–317). J Neurogastroenterol Motil. 17:427–429.
Levasseur A, Merhej V, Baptiste E, Sharma V, Pontarotti P, Raoult D. 2017. The rhizome of Lokiarchaeota illustrates the mosaicity of archaeal genomes. Genome Biol Evol. 9: 2635–2639.
Li Q, Han Y, Dy ABC, Hagerman RJ. 2017. The gut microbiota and autism spectrum disorders. *Front Cell Neurosci*. 11:120.
MacFabe DF, Cain DP, Rodriguez-Capote K, Franklin AE, Hoffman JE, Boon F, Taylor AR, Kavaliers M, Ossenkopp KP.etal.2007. Neurobiological effects of intraventricular propionic acid in rats: possible role of short chain fatty acids on the pathogenesis and characteristics of autism spectrum disorders. *Behav Brain Res*. 176:149–169.
MacFabe DF. 2012. Short-chain fatty acid fermentation products of the gut microbiome: implications in autism spectrum disorders. *Microb Ecol Health Dis*. 23:10.3402.
Manichanh C, Reeder J, Gibert P, Varela E, Llopis M, Antolin M, Guigo R,etal. Knight R, Guarner F. 2010. Reshaping the gut microbiome with bacterial transplantation and antibiotic intake. *Genome Res*. 20:1411–1419.
Margulis L, Chapman MJ. 1998. Endosymbioses: cyclical and permanent in evolution. Trends Microbiol. 6: 342–345, discussion 345–346.
Margulis L, Fester R. 1991. Bellagio Conference and Book. Symbiosis as Source of Evolutionary Innovation: Speciation and Morphogenesis. Conference; June 25–30, 1989; Bellagio Conference Center, Italy. Symbiosis 11. p. 93–101.
Marini E, Maldonado-Contreras AL, Cabras S, Hidalgo G, Buffa R, Marin A, Floris G, Racugno W, Pericchi LR, Castellanos ME, et al. 2007. *Helicobacter pylori* and intestinal parasites are not detrimental to the nutritional status of Amerindians. *Am J Trop Med Hyg*. 76:534–540.
Martín R, Miquel S, Benevides L, Bridonneau C, Robert V, Hudault S, Chain F, Berteau O, Azevedo V, Chatel JM, et al. 2017. Functional characterization of novel *Faecalibacterium prausnitzii* strains isolated from healthy volunteers: a step forward in the use of *F. prausnitzii* as a next-generation probiotic. *Front Microbiol*. 8:1226.
Martinez C, Andres F, Balciunas EM, Manuel Salgado J, Gonzalez D, Manuel J, Converti A, Souza Olivera RP. 2013. Lactic acid properties, applications and production: a review. *Trends Food Sci Technol*. 30:70.
Mazmanian SK, Liu CH, Tzianabos AO, Kasper DL. 2005. An immunomodulatory molecule of symbiotic bacteria directs maturation of the host immune system. Cell. 122: 107–118.
McInerney J, Pisani D, O'Connell MJ. 2015. The ring of life hypothesis for eukaryote origins is supported by multiple kinds of data. Philos Trans R Soc Lond B Biol Sci. 370: 20140323.
McKeown RE. 2009. The epidemiologic transition: changing patterns of mortality and population dynamics. Am J Lifestyle Med. 3:19S–26S.
McSorley HJ, Gaze S, Daveson J, Jones D, Anderson RP, Clouston A, Ruyssers NE, Speare R, McCarthy JS, Engwerda CR, et al. 2011. Suppression of inflammatory immune responses in celiac disease by experimental hookworm infection. *PLoS ONE*. 6:e24092.
Merhej V, Raoult D. 2012. Rhizome of life, catastrophes, sequence exchanges, gene creations, and giant viruses: how microbial genomics challenges Darwin. *Front Cell Infect Microbiol*. 2:113.
Nash AK, Auchtung TA, Wong MC, Smith DP, Gesell JR, Ross MC, Stewart CJ, et al. 2017. The gut mycobiome of the Human Microbiome Project healthy cohort. Microbiome. 5:153.
Nicolucci AC, Hume MP, Martínez I, Mayengbam S, Walter J, Reimer RA. 2017. Prebiotics reduce body fat and alter intestinal microbiota in children who are overweight or with obesity. Gastroenterology. 153:711–722.
O'Sullivan E, Barrett E, Grenham S, Fitzgerald P, Stanton C, Ross RP, Quigley EM, Cryan JF, Dinan TG. 2011. BDNF expression in the hippocampus of maternally separated rats: does *Bifidobacterium breve* 6330 alter BDNF levels? *Benef Microbes*. 2:199–207.
Obregon-Tito AJ, Tito RY, Metcalf J, Sankaranarayanan K, Clemente JC, Ursell LK, Zech Xu Z, Van Treuren W, Knight R, Gaffney PM, et al. 2015. Subsistence strategies in traditional societies distinguish gut microbiomes. *Nat Commun*. 6:6505.
O'Mahony SM, Clarke G, Borre YE, Dinan TG, Cryan JF. 2015. Serotonin, tryptophan metabolism and the brain-gut-microbiome axis. Behav Brain Res. 277:32–48.
Osborn DA, Sinn JK. 2013. Prebiotics in infants for prevention of allergy. *Cochrane Database Syst Rev*.(3): CD006474.

Palma ML, Zamith-Miranda D, Martins FS, Bozza FA, Nimrichter L, Montero-Lomeli M, Marques ETA, Douradinha B. 2015. Probiotic *Saccharomyces cerevisiae* strains as biotherapeutic tools: is there room for improvement? Appl Microbiol Biotechnol. 99:6563–6570.

Parfrey LW, Walters WA, Knight R. 2011. Microbial eukaryotes in the human microbiome: ecology, evolution, and future directions. *Front Microbiol*. 2:153.

Pastell H, Westermann P, Meyer AS, Tuomainen P, Tenkanen M. 2009. In vitro fermentation of arabinoxylan-derived carbohydrates by bifidobacteria and mixed fecal microbiota. J Agric Food Chem. 57:8598–8606.

Paun A, Danska JS. 2015. Immuno-ecology: how the microbiome regulates tolerance and autoimmunity. Curr Opin Immunol. 37:34–39.

Rampelli S, Schnorr SL, Consolandi C, Turroni S, Severgnini M, Peano C, Brigidi P, etal.Crittenden AN, Henry AG, Candela M. 2015. Metagenome sequencing of the Hadza hunter-gatherer gut microbiota. *Curr Biol*. 25: 1682–1693.

Ramulu HG, Raoult D, Pontarotti P. 2012. The rhizome of life: what about metazoa? Front Cell Infect Microbiol. 2: 50.

Raoult D. 2010. The post-Darwinist rhizome of life. Lancet. 375:104–105.

Reddy A, Fried B. 2009. An update on the use of helminths to treat Crohn's and other autoimmunune diseases. Parasitol Res. 104:217–221.

Reynolds LA, Finlay BB, Maizels RM. 2015. Cohabitation in the intestine: interactions among helminth parasites, bacterial microbiota, and host immunity. J Immunol (Baltim MD). 195:4059–4066.

Ricci S, Businaro R, Ippoliti F, Lo Vasco VR, Massoni F, Onofri E, Troili GM, Pontecorvi V, Morelli M, Rapp Ricciardi M, et al. 2013. Altered cytokine and BDNF levels in autism spectrum disorder. *Neurotox Res*. 24: 491–501.

Rios-Covian D, Arboleya S, Hernandez-Barranco AM, Alvarez-Buylla JR, Ruas-Madiedo P, Gueimonde M, de los Reyes-Gavilan CG. 2013. Interactions between bifidobacterium and bacteroides species in cofermentations are affected by carbon sources, including exopolysaccharides produced by bifidobacteria. Appl Environ Microbiol. 79: 7518–7524.

Rivas-Carrillo SD, Pettersson ME, Rubin C-J, Jern P. 2018. Whole-genome comparison of endogenous retrovirus segregation across wild and domestic host species populations. Proc Natl Acad Sci USA. 115:11012–11017.

Rivière A, Moens F, Selak M, Maes D, Weckx S, De Vuyst L. 2014. The ability of bifidobacteria to degrade arabinoxylan oligosaccharide constituents and derived oligosaccharides is strain dependent. Appl Environ Microbiol. 80: 204–217.

Rook GA. 2009. Review series on helminths, immune modulation and the hygiene hypothesis: the broader implications of the hygiene hypothesis. Immunology. 126: 3–11.

Rosenberg E, Zilber-Rosenberg I. 2011. Symbiosis and development: the hologenome concept. Birth Defects Res C Embryo Today. 93:56–66.

Rosenfeld CS. 2015. Microbiome disturbances and autism spectrum disorders. Drug Metab Dispos Biol Fate Chem. 43:1557–1571.

Saad N, Delattre C, Urdaci M, Schmitter JM, Bressollier P. 2013. An overview of the last advances in probiotic and prebiotic field. LWT - Food Sci Technol. 50:1–16.

Salvucci E. 2012. Selfishness, warfare, and economics; or integration, cooperation, and biology. Front Cell Infect Microbiol. 2:54.

Salvucci E. 2013. Consequences of biome depletion. Acta Biol Colomb. 18:31–42.

Salvucci E. 2016. Microbiome, holobiont and the net of life. Crit Rev Microbiol. 42:485–494.

Samaco RC, Neul JL. 2011. Complexities of Rett syndrome and MeCP2. J Neurosci. 31:7951–7959.

Sanchez JI, Marzorati M, Grootaert C, Baran M, Van Craeyveld V, Courtin CM, Broekaert WF, Delcour JA, Verstraete W, Van de Wiele T. 2009. Arabinoxylan-oligosaccharides (AXOS) affect the protein/carbohydrate fermentation balance and microbial population dynamics of the Simulator of Human Intestinal Microbial Ecosystem. *Microb Biotechnol*. 2:101–113.

Sandiego CM, Gallezot J-D, Pittman B, Nabulsi N, Lim K, Lin S-F, Matuskey D, Lee JY, O'Connor KC, Huang Y, et al. 2015. Imaging robust microglial activation after lipopolysaccharide administration in humans with PET. *Proc Natl Acad Sci USA*. 112:12468–12473.

Savignac HM, Corona G, Mills H, Chen L, Spencer JPE, Tzortzis G, Burnet PWJ. 2013. Prebiotic feeding elevates central brain derived neurotrophic factor, N-methyl-D-aspartate receptor subunits and D-serine. Neurochem Int. 63:756–764.

Savignac HM, Kiely B, Dinan TG, Cryan JF. 2014. Bifidobacteria exert strain-specific effects on stress-related behavior and physiology in BALB/c mice. Neurogastroenterol Motil. 26:1615–1627.

Savinov AB. 2012. Autocenosis and democenosis as symbiotic systems and biological notions. Zhurnal Obshcheĭ Biol. 73:284–301.

Schneider KM, Wirtz TH, Kroy D, Albers S, Neumann UP, Strowig T, Sellge G, Trautwein C. 2018. Successful fecal microbiota transplantation in a patient with severe complicated *Clostridium difficile* infection after liver transplantation. CRG. 12:76–84.

Schnorr SL, Candela M, Rampelli S, Centanni M, Consolandi C, Basaglia G, Turroni S, Biagi E, Peano C, Severgnini M, et al. 2014. Gut microbiome of the Hadza hunter-gatherers. *Nat Commun*. 5:3654.

Schnorr SL, Sankaranarayanan K, Lewis CM, Warinner C. 2016. Insights into human evolution from ancient and contemporary microbiome studies. Curr Opin Genet Dev. 41:14–26.

Schroeder FA, Lin CL, Crusio WE, Akbarian S. 2007. Antidepressant-like effects of the histone deacetylase inhibitor, sodium butyrate, in the mouse. Biol Psychiatry. 62:55–64.

Sender R, Fuchs S, Milo R. 2016. Revised estimates for the number of human and bacteria cells in the body. *PLoS Biol*. 14:e1002533.

Severance EG, Yolken RH, Eaton WW. 2016. Autoimmune diseases, gastrointestinal disorders and the microbiome in

schizophrenia: more than a gut feeling. *Schizophr Res*. 176:23–35.
Singh Y, Ahmad J, Musarrat J, Ehtesham NZ, Hasnain SE. 2013. Emerging importance of holobionts in evolution and in probiotics. Gut Pathog. 5:12.
Sleator RD. 2010. The human superorganism – of microbes and men. Med Hypoth. 74:214–215.
Smits SA, Leach J, Sonnenburg ED, Gonzalez CG, Lichtman JS, Reid G, Knight R, Manjurano A, Changalucha J, Elias JE, et al. 2017. Seasonal cycling in the gut microbiome of the Hadza hunter-gatherers of Tanzania. *Science*. 357: 802–806.
Sonnenburg ED, Smits SA, Tikhonov M, Higginbottom SK, Wingreen NS, Sonnenburg JL. 2016. Diet-induced extinctions in the gut microbiota compound over generations. Nature. 529:212–215.
Sonnenburg ED, Sonnenburg JL. 2014. Starving our microbial self: the deleterious consequences of a diet deficient in microbiota-accessible carbohydrates. Cell Metab. 20: 779–786.
Strati F, Cavalieri D, Albanese D, Felice CD, Donati C, Hayek J, Jousson O, Leoncini S, Renzi D, Calabrò A, et al. 2017. New evidences on the altered gut microbiota in autism spectrum disorders. *Microbiome*. 5:24.
Summers RW, Elliott DE, Urban JF, Thompson RA, Weinstock JV. 2005a. *Trichuris suis* therapy for active ulcerative colitis: a randomized controlled trial. Gastroenterology. 128:825–832.
Summers RW, Elliott DE, Urban JF, Thompson R, Weinstock JV. 2005b. *Trichuris suis* therapy in Crohn's disease. Gut. 54:87–90.
Theoharides TC, Athanassiou M, Panagiotidou S, Doyle R. 2015. Dysregulated brain immunity and neurotrophin signaling in Rett syndrome and autism spectrum disorders. J Neuroimmunol. 279:33–38.
Thomas S, Metzke D, Schmitz J, Dörffel Y, Baumgart DC. 2011. Anti-inflammatory effects of *Saccharomyces boulardii* mediated by myeloid dendritic cells from patients with Crohn's disease and ulcerative colitis. Am J Physiol Gastrointest Liver Physiol. 301:G1083–G1092.
Tilg H, Kaser A. 2011. Gut microbiome, obesity, and metabolic dysfunction. J Clin Invest. 121:2126–2132.
Tito RY, Knights D, Metcalf J, Obregon-Tito AJ, Cleeland L, Najar F, Roe B, Reinhard K, Sobolik K, Belknap S, et al. 2012. Insights from characterizing extinct human gut microbiomes. *PLoS ONE*. 7:e51146.
Tristem M. 2000. Identification and characterization of novel human endogenous retrovirus families by phylogenetic screening of the human genome mapping project database. J Virol. 74:3715–3730.
Truchado P, Hernandez-Sanabria E, Salden BN, Van den Abbeele P, Vilchez-Vargas R, Jauregui R, Pieper DH, Possemiersa S, Van de Wiele T. 2017. Long chain arabinoxylans shift the mucosa-associated microbiota in the proximal colon of the simulator of the human intestinal microbial ecosystem (M-SHIME). *J Funct Foods*. 32: 226–237.
Turroni S, Fiori J, Rampelli S, Schnorr SL, Consolandi C, Barone M, Biagi E, Fanelli F, Mezzullo M, Crittenden A, et al. 2016. Fecal metabolome of the Hadza hunter-gatherers: a host-microbiome integrative view. *Sci Rep*. 6: 32826.
van der Meulen TA, Harmsen HJM, Bootsma H, Spijkervet FKL, Kroese FGM, Vissink A. 2016. The microbiome systemic diseases connection. *Oral Dis*.73:14741–14750.
van Zanten GC, Knudsen A, Röytiö H, Forssten S, Lawther M, Blennow A, Lahtinen SJ, Jakobsen M, Svensson B, Jespersen L. 2012. The effect of selected synbiotics on microbial composition and short-chain fatty acid production in a model system of the human colon. *PLoS One*. 7: e47212.
Vandenplas Y, Huys G, Daube G. 2015. Probiotics: an update. J Pediatr (Rio J). 91:6–21.
Vangay P, Johnson AJ, Ward TL, Al-Ghalith GA, Shields-Cutler RR, Hillmann BM, Lucas SK, Beura LK, Thompson EA, Till LM, et al. 2018. US immigration westernizes the human gut microbiome. *Cell*. 175: 962–972.e10.
Vannier-Santos MA, Lenzi HL. 2011. Parasites or cohabitants: cruel omnipresent usurpers or creative "éminences grises"? J Parasitol Res. 2011:214174.
Weinstock JV, Elliott DE. 2013. Translatability of helminth therapy in inflammatory bowel diseases. Int J Parasitol. 43:245–251.
Welters CFM, Heineman E, Thunnissen FBJM, van den Bogaard AEJM, Soeters PB, Baeten CGMI. 2002. Effect of dietary inulin supplementation on inflammation of pouch mucosa in patients with an ileal pouch-anal anastomosis. Dis Colon Rectum. 45:621–627.
West CE, Jenmalm MC, Prescott SL. 2015. The gut microbiota and its role in the development of allergic disease: a wider perspective. Clin Exp Allergy J Br Soc Allergy Clin Immunol. 45:43–53.
Xu X, Miller EC, Pozzo-Miller L. 2014. Dendritic spine dysgenesis in Rett syndrome. Front Neuroanat. 8:97.
Yatsunenko T, Rey FE, Manary MJ, Trehan I, Dominguez-Bello MG, Contreras M, Magris M, Hidalgo G, Baldassano RN, Anokhin AP, et al. 2012. Human gut microbiome viewed across age and geography. *Nature*. 486:222–227.